Technical Report
Forensic File Signature Analysis

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Group 1

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I. INTRODUCTION

The forensic file signature analysis is more about the data hiding forensics. It is a process whose job is to match the suspected data type with the database. Many tools could perform this comparison process. The tools used in this report are HxD Hex Editorin and HexBrowser in Windows OS. This report aims to introduce the analysis process of the file signature, how to find the correct application to associate them, and the recovery process.

II. PROCEDURE

1. Manually Analyze the file signatures with HxD Hex Editorin in Windows OS

Every file has a unique hex signature and will identify the type of the particular file regardless of changing, deleting, or hiding the extension. To figure out its appropriate file extension File signatures to aid with the restoration process of the unreadable files. Signature analysis is a process where file headers and extensions are compared with a known database of file headers and extensions to verify all files on the storage media and discover those that may be hidden. So, we'll manually analyze the header and trailer of these files by HEX Editor to identify their type.

1) Try to hide a confidential PDF file as JPG and stored it in a folder of pictures.

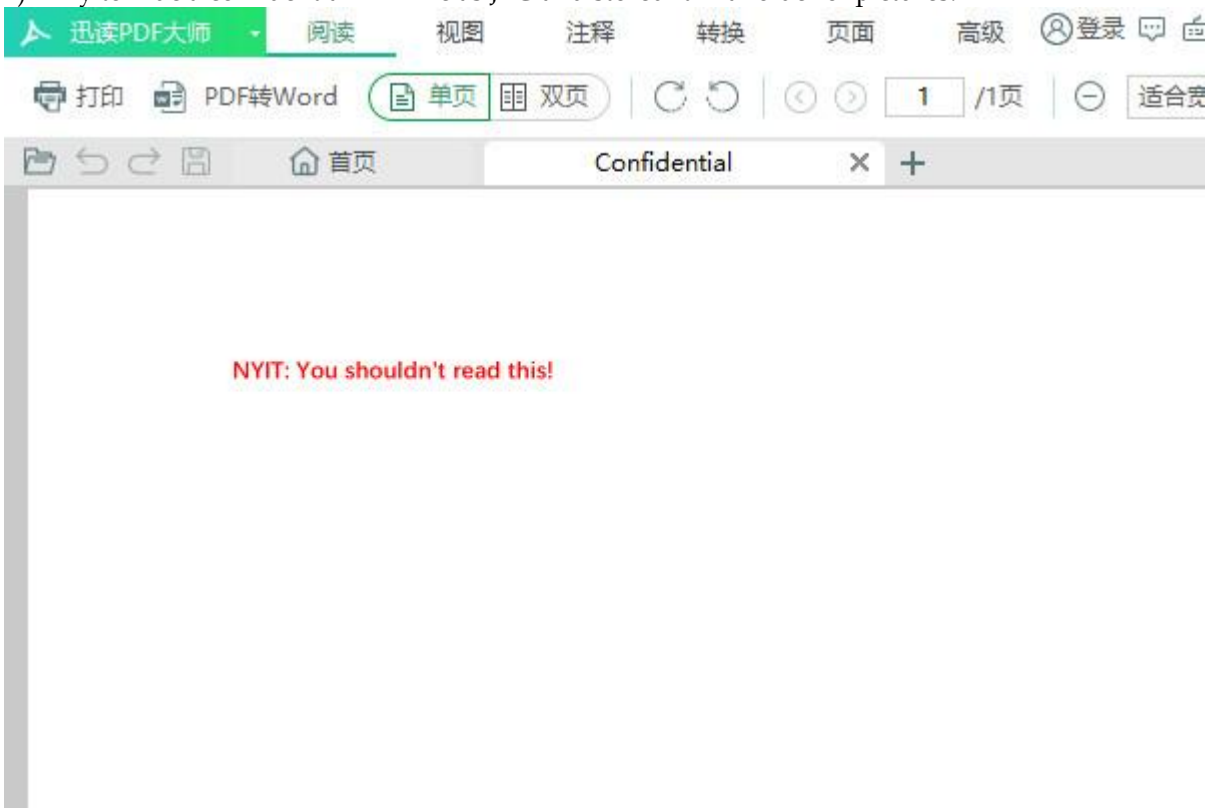


Figure 1

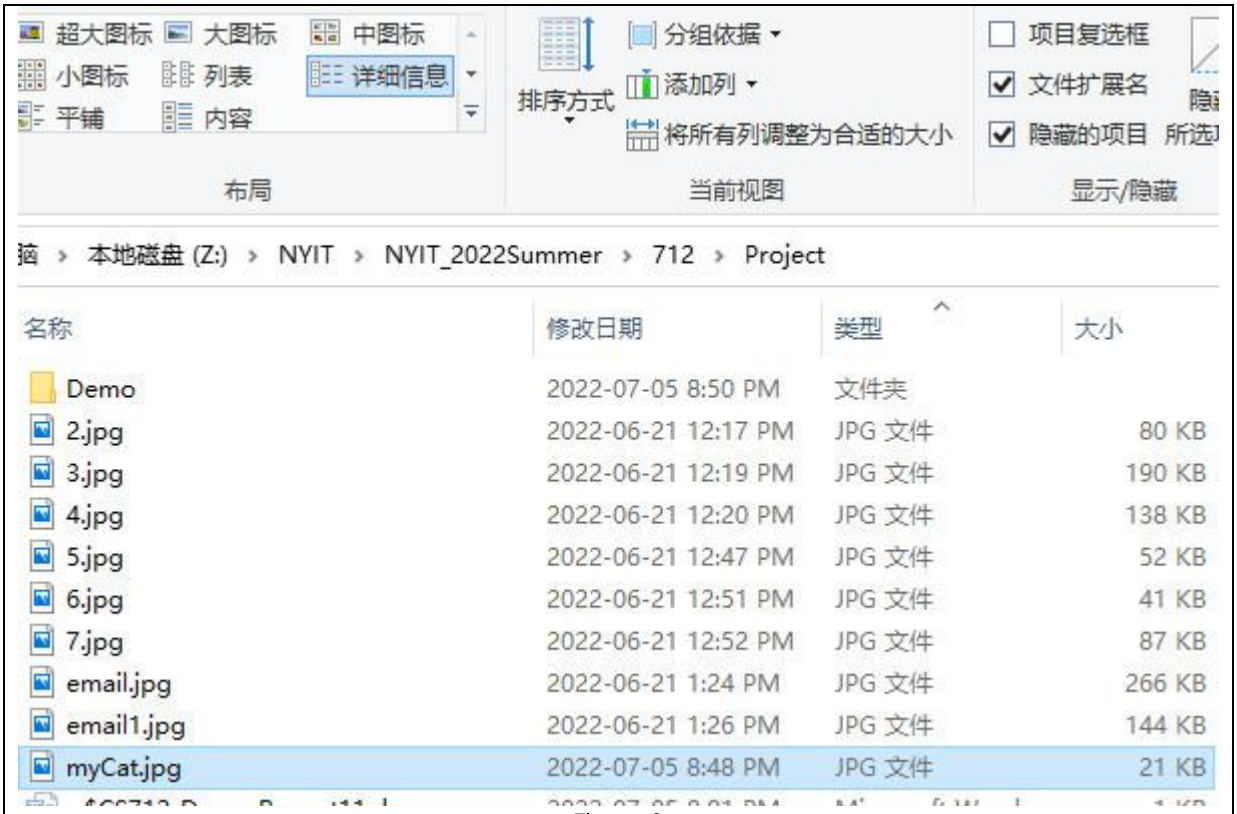


Figure 2

2) This myCat.jpg can't open by media application. Then, open the file in HxD Hex Editorin shown in Figure 4, and we can find the header.

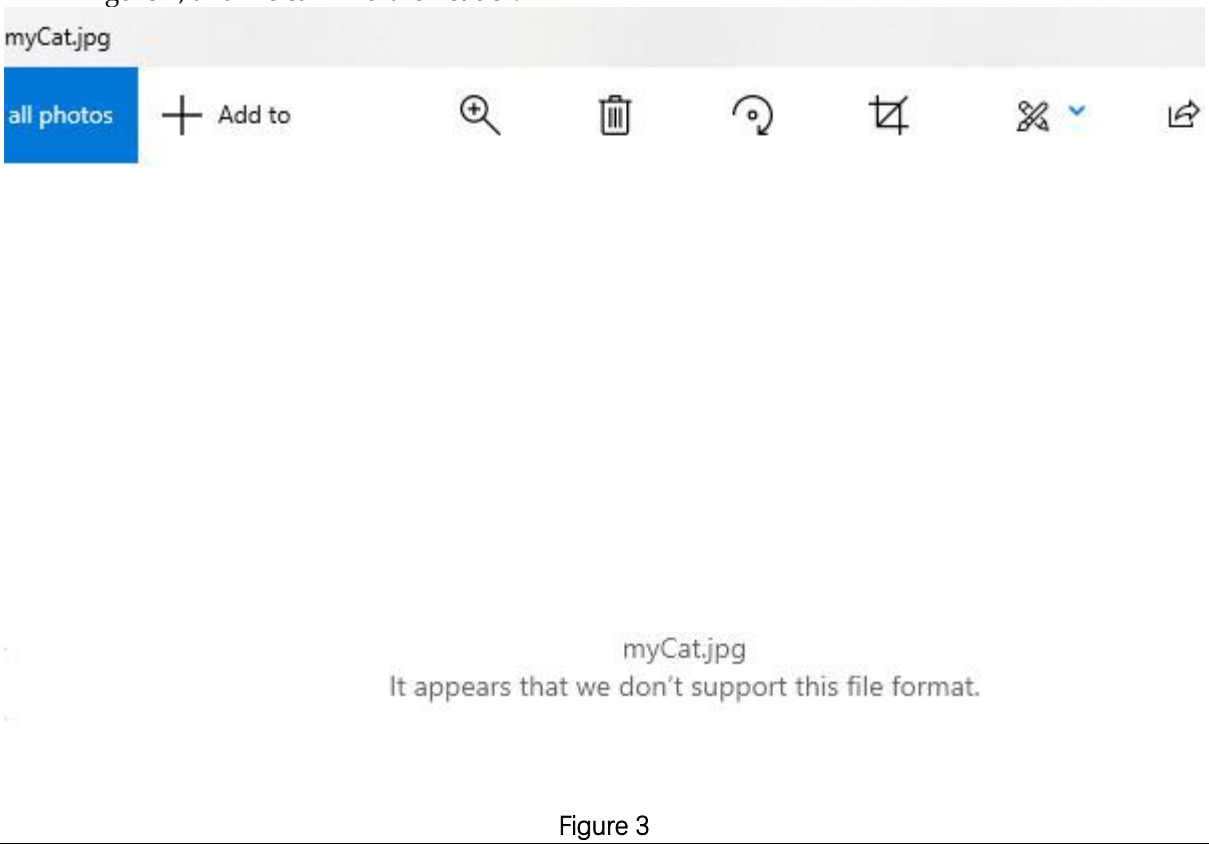


Figure 3

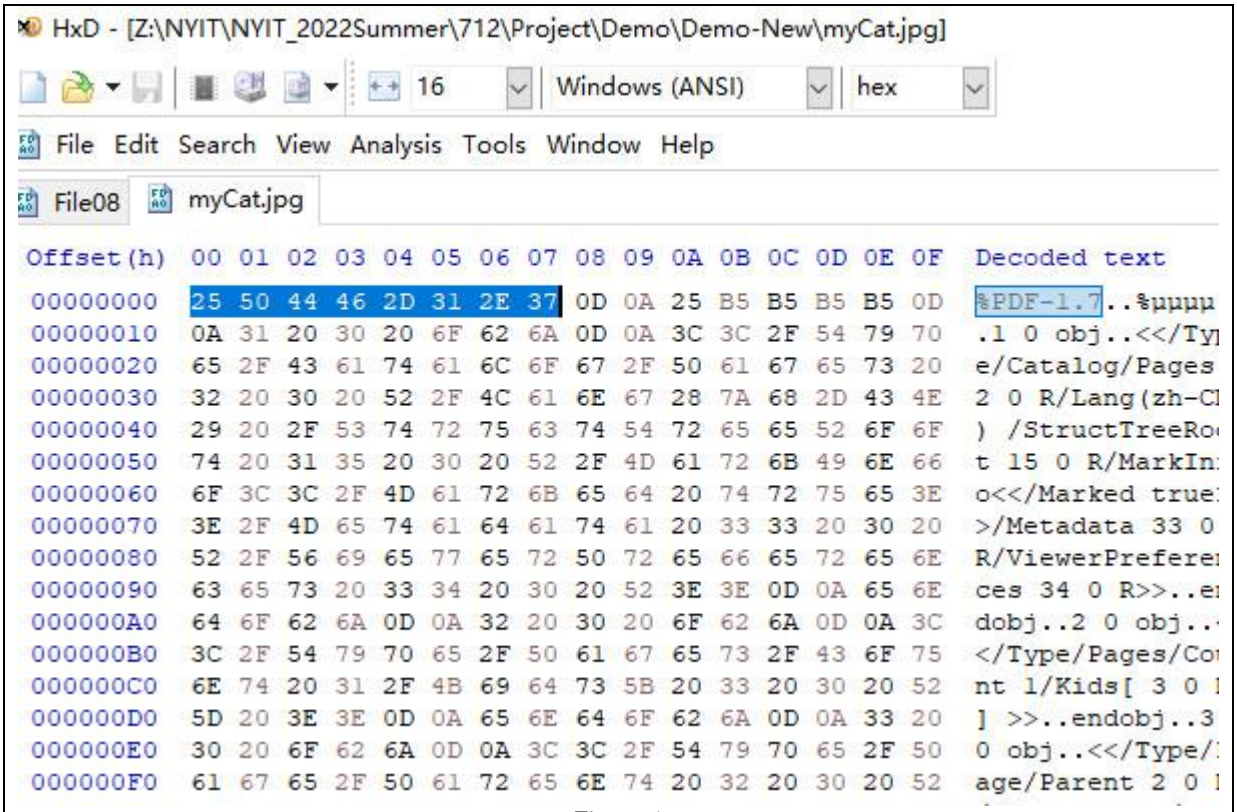


Figure 4

3) Copy the group of bytes highlighting over there and go to the website to find its associate file extension.

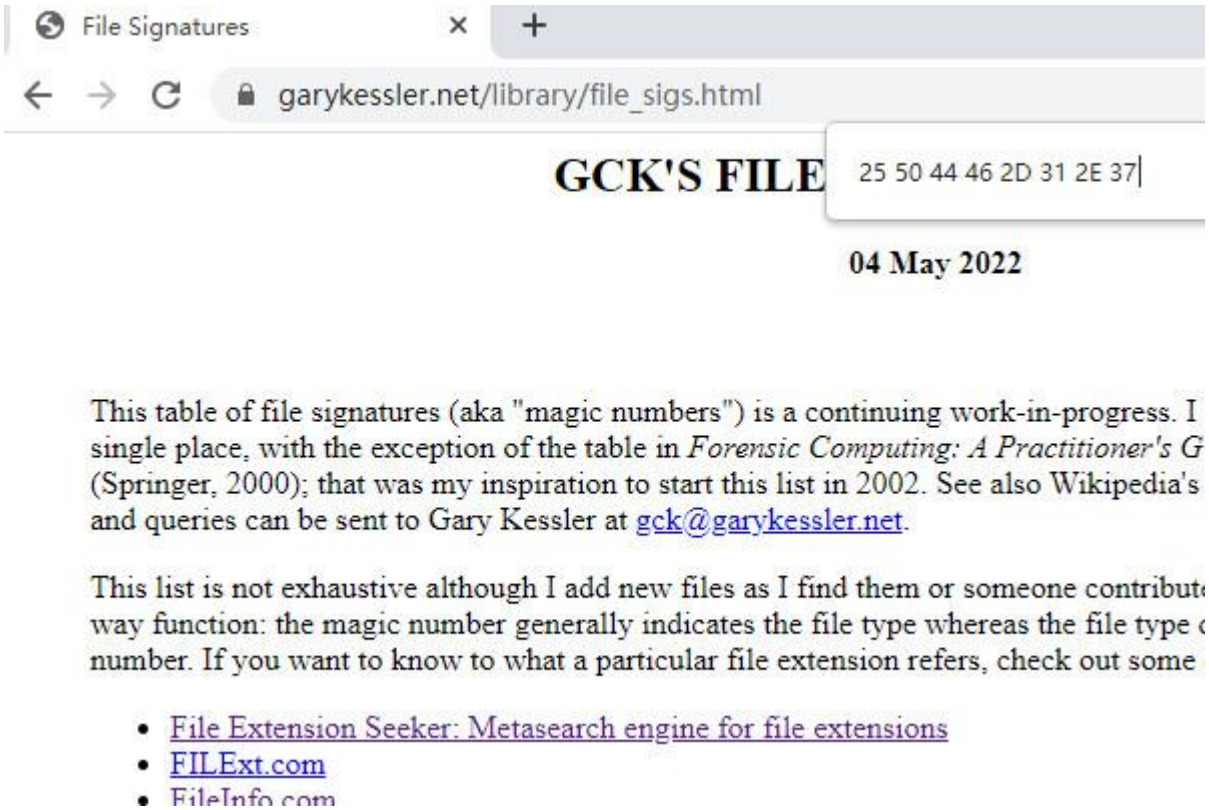


Figure 5

4) Start backspacing until it finds results. In Figure 6, the four bytes are known as pdf format, and its file description is the same as in HxD Hex Editorin in Figure 4. Additionally, we compare the trailers at the end of the hex shown in Figure 7.

```
25 21 50 53 2D 41 64 6F     %!PS-Ado
62 65 2D                    be-
                        PS  PostScript file

25 21 50 53 2D 41 64 6F     %!PS-Ado
62 65 2D 33 2E 30 20 45     be-3.0 E
50 53 46 2D 33 20 30        PSF-3.0
                       EPS  Adobe encapsulated PostScript file
                            (If this signature is not at the immediate
                            beginning of the file, it will occur early
                            in the file, commonly at byte offset 30 [0x1E])

25 50 44 46                 %PDF
                PDF, FDF, AI Adobe Portable Document Format, Forms Documen
                            Trailers:
                            0A 25 25 45 4F 46(.%%EOF)
                            0A 25 25 45 4F 46 0A(.%%EOF.)
                            0D 0A 25 25 45 4F 46 0D 0A(..%%EOF.
                            0D 25 25 45 4F 46 0D(.%%EOF.)
                            NOTE: There may be multiple end-of-file marks wi
                            file. When carving, be sure to get the last one.
```

Figure 6

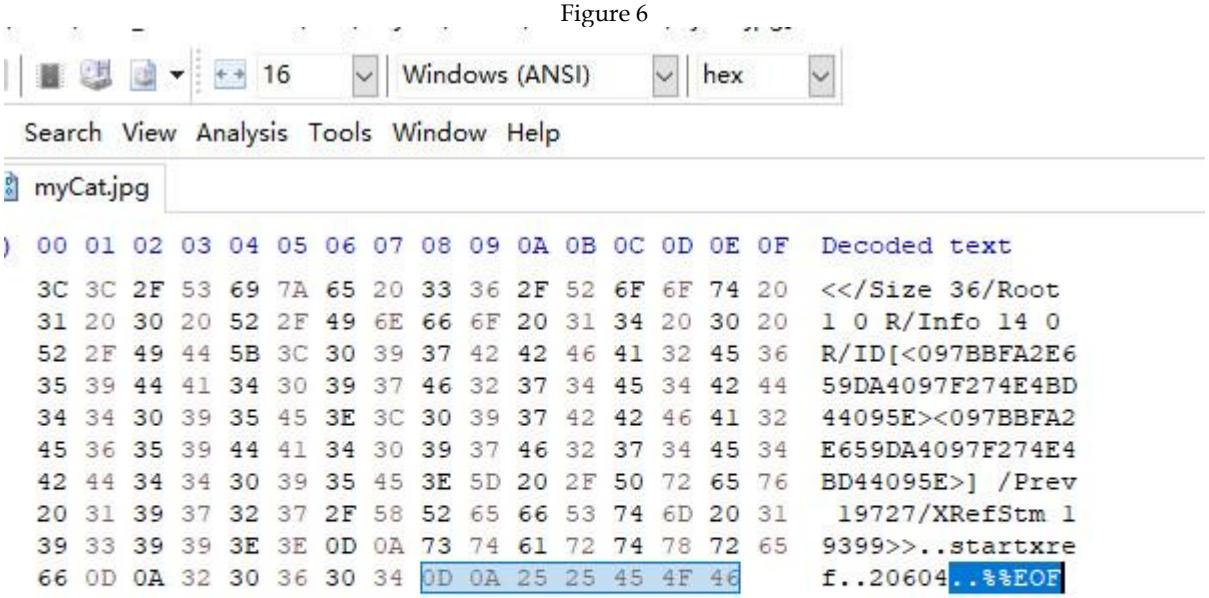


Figure 7

5) Change the file extension to ".pdf" and try to open it in word.

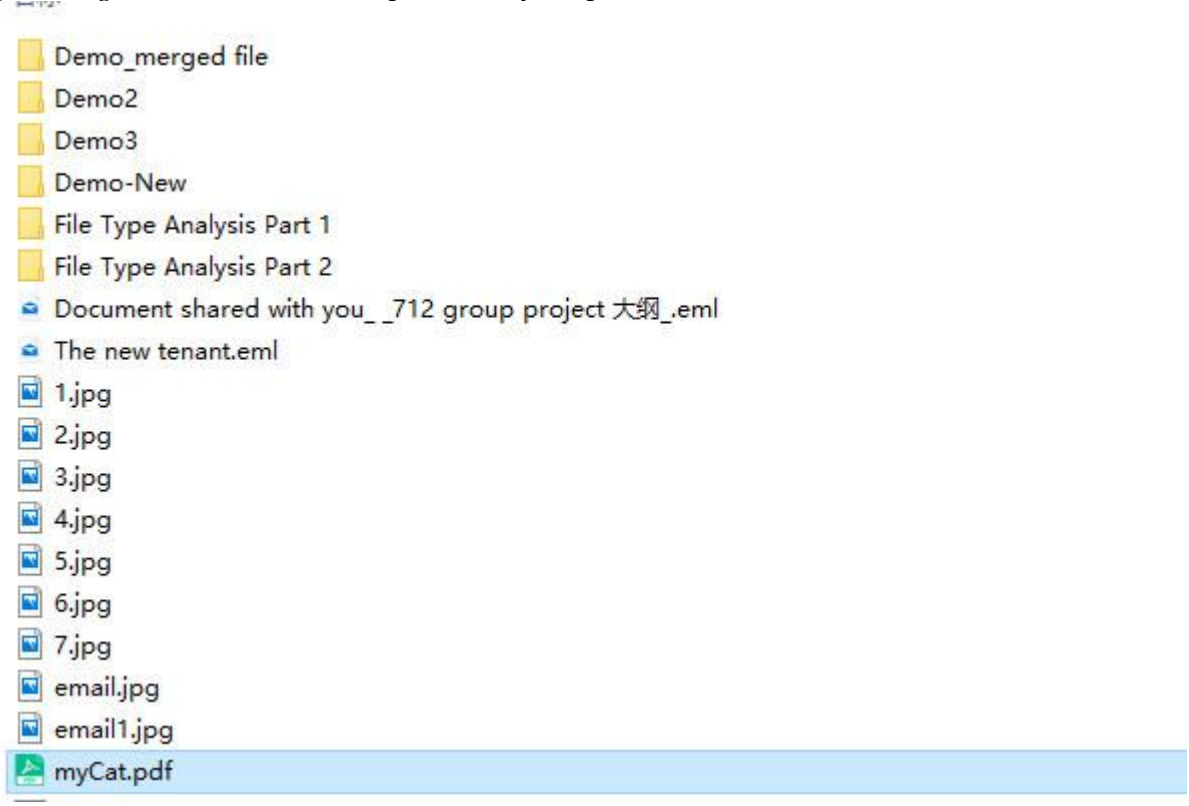


Figure 8

6) Compare the result in both pdf and in HxD Hex Editorin.

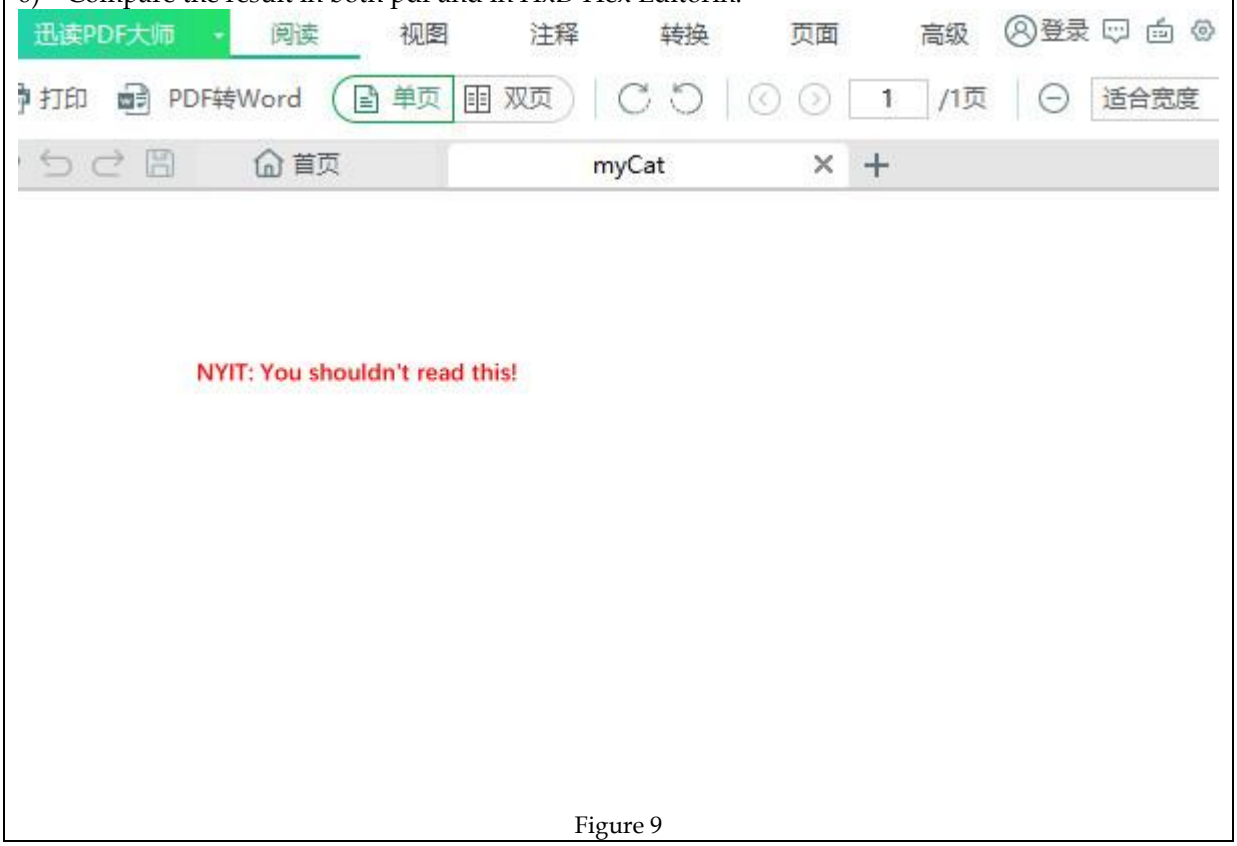


Figure 9

2. Automatically Analyze the file signatures with HexBrowser in Windows OS

HexBrowser is a freeware that identifies file types. It now recognizes more than 1000 file formats by looking for signatures inside the files. Hence, we don't need to determine the file's extension by checking every file's signature. We will automate the analysis process using HexBrowser in Windows OS in this part.

1) Open the HexBrowser and choose the file that needs to be analyzed. We can find the file type shown in Figure 10.

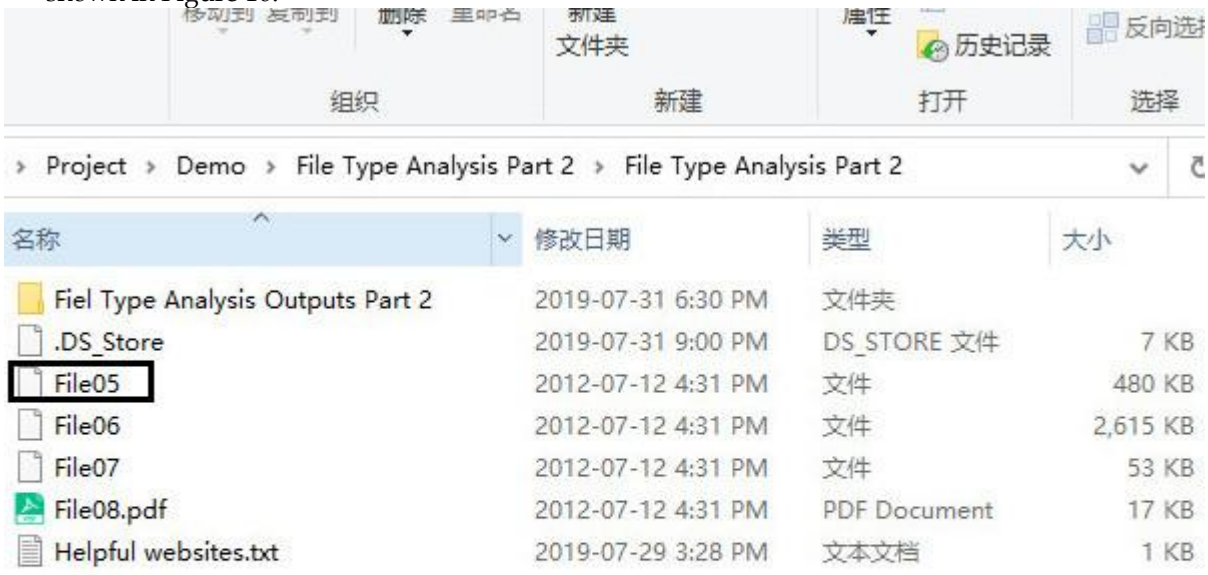


Figure 10

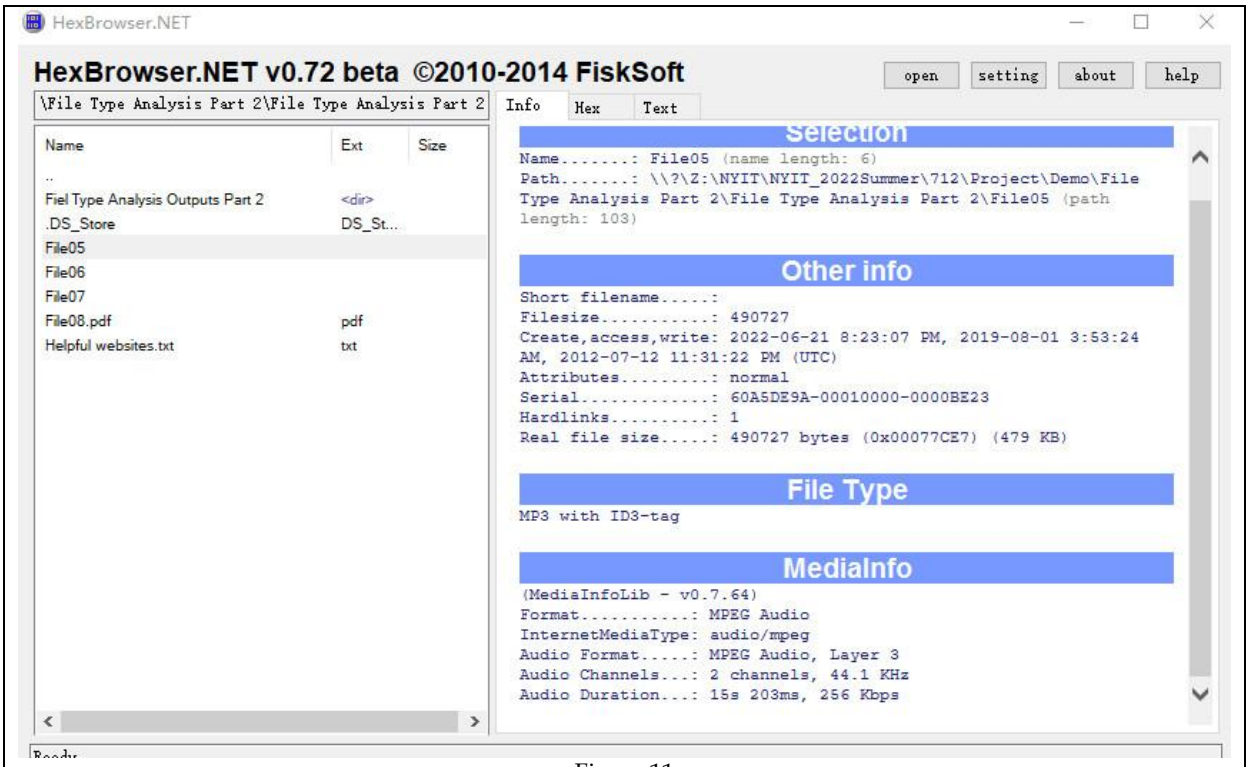


Figure 11

2) Change the file extension to ".mp3" and try to open it in any audio application.

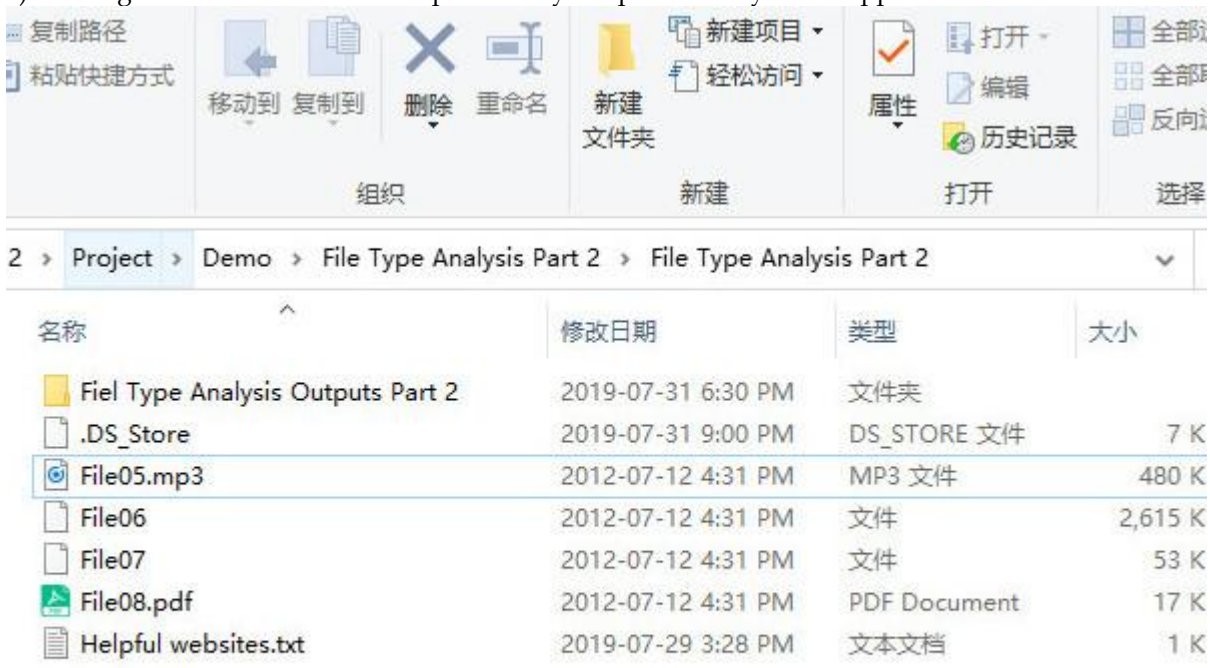


Figure 12

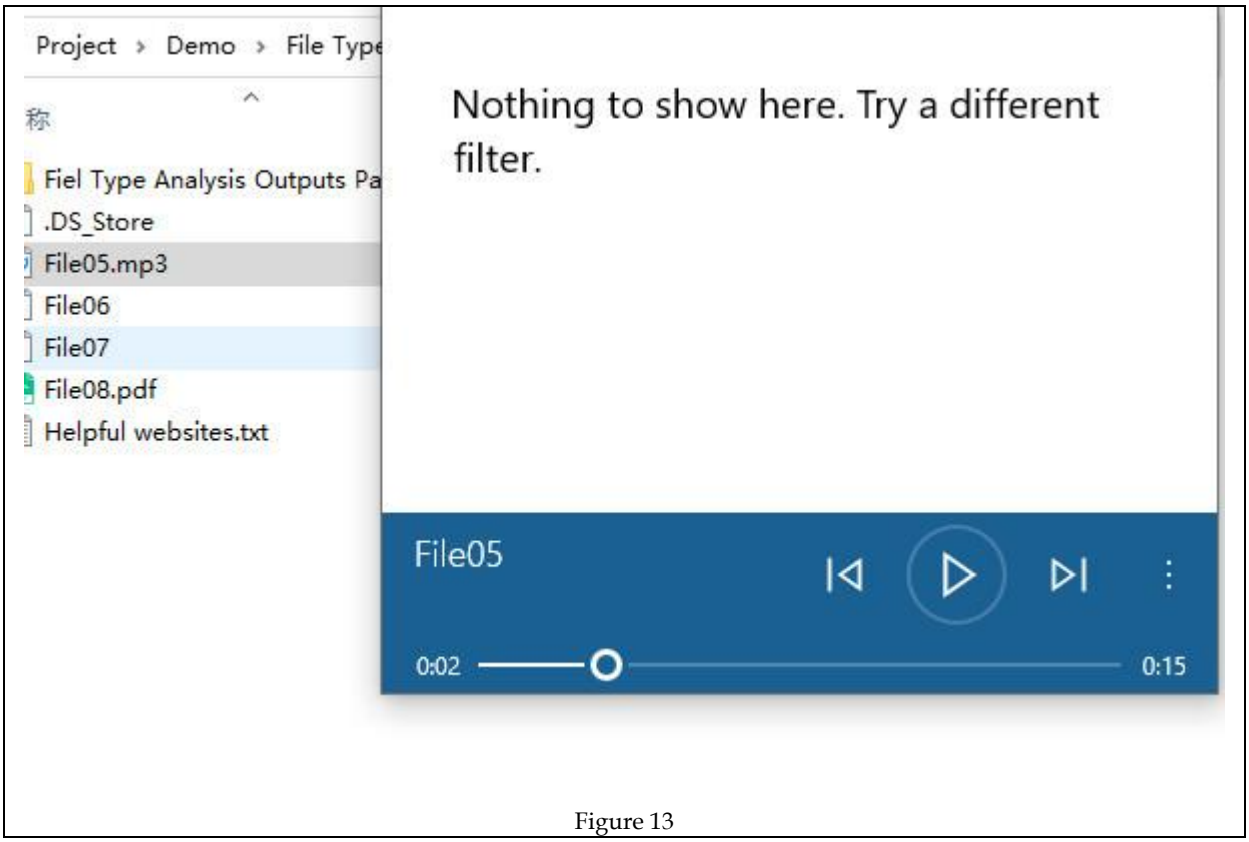

Project › Demo › File Type
称
Fiel Type Analysis Outputs Pa
.DS_Store
File05.mp3
File06
File07
File08.pdf
Helpful websites.txt
Nothing to show here. Try a different filter.
File05
0:02
0:15

Figure 13

Reference

[1] Nihad Ahmad Hassan and Rami Hijazi. 2016. *Data Hiding Techniques in Windows OS: A Practical Approach to Investigation and Defense (1st. ed.)*. Syngress Publishing.